Machine Learning – Assignment 2

M.Tech (AIML), BITS Pilani WILP
Your Name (20XXAAXXXXX)
Dataset: UCI Dry Bean Dataset (ID 602)

1. GitHub Repository Link

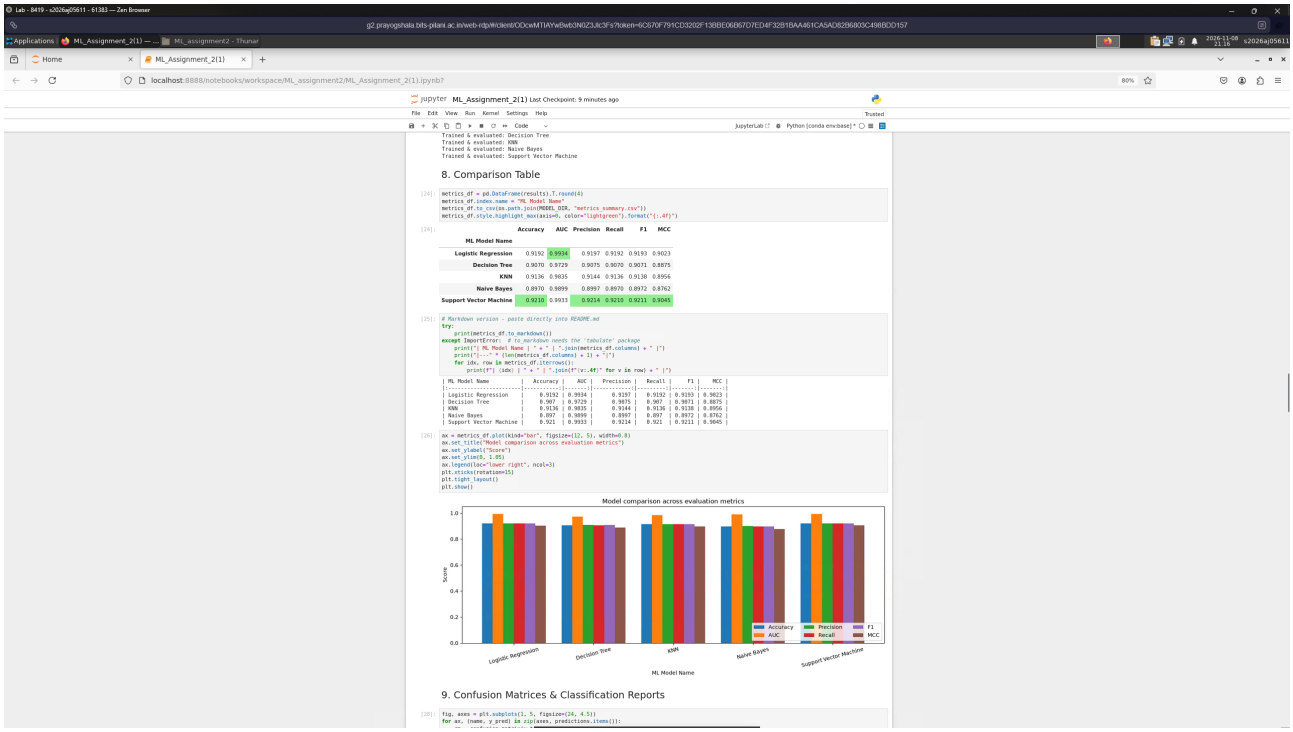
https://github.com/<username>/<repo>

Contains complete source code, requirements.txt, README.md, test_data.csv and model/ (saved models for all five classifiers).

2. Live Streamlit App Link

https://<your-app>.streamlit.app

3. Screenshot – Execution on BITS Virtual Lab



4. README Content

a. Problem statement

Dry beans of different varieties look similar, and sorting them by hand is slow and inconsistent. The goal is to predict the variety of a dry bean grain (one of 7 classes) from 16 shape features extracted from its image. Five classifiers - Logistic Regression, Decision Tree, K-Nearest Neighbors, Gaussian Naive Bayes and Support Vector Machine - are trained on the same data, compared on Accuracy, AUC, Precision, Recall, F1 and MCC, and served through an interactive Streamlit app.

b. Dataset description

Source: UCI Machine Learning Repository - Dry Bean Dataset (ID 602), https://archive.ics.uci.edu/dataset/602/dry+bean+dataset. Images of 13,611 grains of 7 registered dry bean varieties were captured with a high-resolution camera, and 16 features (12 dimensional and 4 shape-form features) were extracted from each grain. After removing 68 duplicate rows, 13,543 instances were used. There are no missing values and all features are numeric. The data was split 80/20 (stratified, random_state=42): 10,834 training rows and 2,709 test rows (saved as test_data.csv). Features were median-imputed and standardised inside each model pipeline. Because this is a 7-class problem, AUC is one-vs-rest (weighted) and Precision/Recall/F1 are weighted averages.

Class	Count
DERMASON	3546
SIRA	2636
SEKER	2027
HOROZ	1860
CALI	1630
BARBUNYA	1322
BOMBAY	522

Feature	Description
Area	Area of the bean zone, in pixels
Perimeter	Length of the bean's border
MajorAxisLength	Length of the longest line that can be drawn through the bean
MinorAxisLength	Longest line perpendicular to the major axis
AspectRation	Ratio of major to minor axis length
Eccentricity	Eccentricity of the ellipse with the same moments as the region
ConvexArea	Pixels in the smallest convex polygon containing the bean
EquivDiameter	Diameter of a circle with the same area as the bean
Extent	Ratio of bean pixels to bounding-box pixels
Solidity	Ratio of bean pixels to convex-hull pixels
roundness	4 * pi * Area / Perimeter^2
Compactness	EquivDiameter / MajorAxisLength
ShapeFactor1	Shape factor 1
ShapeFactor2	Shape factor 2
ShapeFactor3	Shape factor 3
ShapeFactor4	Shape factor 4

c. GitHub Repository Link

https://github.com/<username>/<repo>

```
project-folder/
|-- app.py                    # Streamlit app
|-- ML_Assignment_2.ipynb     # training, evaluation and file generation
|-- requirements.txt
|-- README.md
|-- test_data.csv             # held-out test split used in the experiments
|-- data/Dry_Bean_Dataset.csv  # full dataset (CSV copy)
|-- .streamlit/config.toml    # app theme
|-- model/
    |-- logistic_regression.pkl
    |-- decision_tree.pkl
    |-- knn.pkl
    |-- naive_bayes.pkl
    |-- support_vector_machine.pkl
    |-- label_encoder.pkl
    |-- metadata.json
    |-- metrics_summary.csv
```

d. Models used – comparison table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9192	0.9934	0.9197	0.9192	0.9193	0.9023
Decision Tree	0.9070	0.9729	0.9075	0.9070	0.9071	0.8875
KNN	0.9136	0.9835	0.9144	0.9136	0.9138	0.8956
Naive Bayes	0.8970	0.9899	0.8997	0.8970	0.8972	0.8762
Support Vector Machine	0.9210	0.9933	0.9214	0.9210	0.9211	0.9045

Observations

ML Model Name	Observation about model performance
Logistic Regression	TODO
Decision Tree	TODO
KNN	TODO
Naive Bayes	TODO
Support Vector Machine	TODO
Overall, Winner for your dataset?	TODO